

Data Science & AI Bootcamp – Interview Question Bank

MACHINE LEARNING

1. Explain supervised vs unsupervised learning.
2. What is the difference between classification and regression?
3. What is the benefit of using Mean Squared Error (MSE) as a cost function in linear regression compared to Mean Absolute Error (MAE)?
4. Explain different evaluation metrics.
5. How do you handle missing data?
6. Explain feature scaling methods.
7. What is the bias-variance tradeoff?
8. Explain decision trees.
9. What is random forest?
10. How do you handle categorical variables?
11. Explain k-means clustering.
12. What benefits DBSCAN offers over k-means?
13. What is feature engineering?
14. What is feature selection? How does feature selection impact machine learning models?
15. Explain logistic regression.
16. What is ensemble learning and why is it useful?
17. How do you handle imbalanced datasets?
18. Explain bagging and boosting.
19. What is regularization?
20. How do you choose the right ML algorithm?
21. Explain different boosting algorithms.
22. What are support vector machines?
23. Explain dimensionality reduction techniques.
24. What are different clustering algorithms?
25. How do you handle multicollinearity?
26. Explain ROC and AUC curves.
27. What are ensemble stacking methods?
28. How do you handle concept drift?
29. What are semi-supervised learning techniques?
30. How do you handle high-dimensional data?
31. What are different optimization algorithms?
32. What are precision and recall? How do they differ?
33. What is overfitting in machine learning?
34. Explain the concept of cross-validation.

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- 35. What are hyperparameters, and how do they differ from model parameters?
- 36. Discuss transfer learning and scenarios for its application.
- 37. Interpret a ROC curve in relation to model performance.
- 38. What is a training set and a test set in machine learning?
- 39. Summarize how decision trees function along with their advantages and disadvantages.